A More Generated Images

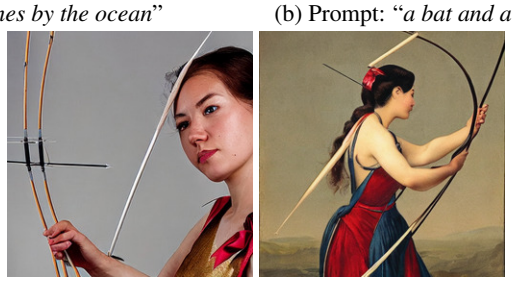
In this section we provide a selection of additional images generated during our experiments.

A.1 Homonym Duplication

Fig. 7 displays further examples of homonym duplication seen in Stable Diffusion.

(a) Prompt: "*tall cranes by the ocean*"

(b) Prompt: "*a bat and a baseball fly through the air*"



(c) Prompt: "*a woman with a silk bow and arrow*"

Figure 7: Examples of homonym duplication

A.2 Summing Encodings

Here we include further examples of images generated from the weighted sum of two prompt encodings as described in §3.

A.2.1 Both Concepts Realised

In Fig. 8 we include examples of images where both concepts in the sum are realised in the final image.

A.2.2 Leakage

In Fig. 9 we include some images that do not fully realise both concepts, but instead allow some kind of concept "*leakage*" as a result of the sum. For example, the prompts "*a completely black cat*" and "*a completely white cat*", both overwhelmingly generate images of cats of one colour, as described. The weighted sum of the two does not generate one completely white cat and one completely black cat. Instead it generates a cat that is both black and white.

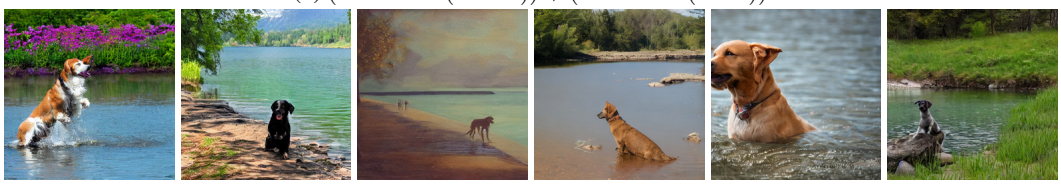
Similarly, the sum of "*a wall painted red*" and "*a wall painted blue*" does not generate a red wall and a blue wall, or even (for the majority of the time) a wall that is both red and blue. Instead, it tends to generate a wall that is one of the two colours described, but where something else in the image (such as a door) is of the other colour.

(a) $(0.5 \cdot \text{CLIP}(\text{“}\textit{elephant}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{snow}\text{”}))$

(b) $(0.5 \cdot \text{CLIP}(\text{“}\textit{goat}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{crown}\text{”}))$

(c) $(0.5 \cdot \text{CLIP}(\text{“}\textit{tiger}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{glasses}\text{”}))$

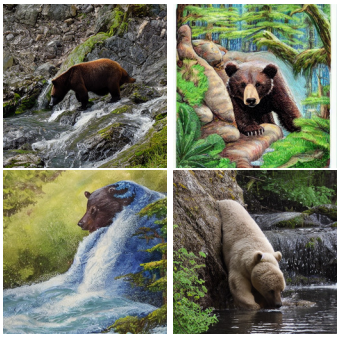
(d) $(0.5 \cdot \text{CLIP}(\text{“}\textit{bear}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{hat}\text{”}))$



(e) $(0.5 \cdot \text{CLIP}(\text{“}\textit{dog}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{lake}\text{”}))$

(f) $(0.5 \cdot \text{CLIP}(\text{“}\textit{cat}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{tree}\text{”}))$

(g) $(0.5 \cdot \text{CLIP}(\text{“}\textit{giraffe}\text{”}))$
$+(0.5 \cdot \text{CLIP}(\text{“}\textit{beach}\text{”}))$



(h) $(0.5 \cdot \text{CLIP}(\text{“}\textit{bear}\text{”}))$
$+(0.5 \cdot \text{CLIP}(\text{“}\textit{waterfall}\text{”}))$

(i) $(0.5 \cdot \text{CLIP}(\text{“}\textit{snow}\text{”}))$
$+(0.5 \cdot \text{CLIP}(\text{“}\textit{church}\text{”}))$

Figure 8: Images generated from sums of encodings demonstrating the realisation of both concepts in the sum

(a) $(0.5 \cdot \text{CLIP}(\text{“}\textit{a completely black cat}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{a completely white cat}\text{”}))$

(b) $(0.5 \cdot \text{CLIP}(\text{“}\textit{a wall painted red}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{a wall painted blue}\text{”}))$

Figure 9: Images generated from sums of encodings that demonstrate concept leakage

A.3 Sense Editing

In this section we provide more example images demonstrating the effects of our linear subspace procedure as described in §4.

For each prompt we display 15 images sampled from the image generated from the unedited prompt encoding, alongside 15 generated from the encoding after editing it to favour each sense.